CMS Physics Analysis Summary

Contact: cms-pag-conveners-higgs@cern.ch 2011/12/13

Search for a Higgs boson decaying into two photons in the CMS detector

The CMS Collaboration

Abstract

This note describes the search for a Higgs boson decaying into diphotons in pp collisions at the LHC at a center-of-mass energy of 7 TeV. Data-driven methods are used to evaluate the reconstruction and selection performance for the signal, and the background is evaluated from a fit to the data. The analysis is performed on a dataset corresponding to 4.8 fb^{-1} of data recorded in 2011 by the CMS experiment. Limits are set on the cross section of a Standard Model Higgs boson decaying to two photons. The expected exclusion limit at 95% CL is between 1.5 and 2.0 times the Standard Model cross section in the mass range between 110 and 140 GeV/c^2, and the observed limit disfavours at 95% CL a Standard Model Higgs boson decaying into two photons in the mass range 127 to 131 GeV/c^2.

1 Introduction

The Standard Model of particle physics (SM) has been very successful in explaining experimental data [1, 2]. A question that remains open is the origin of the masses of the W and Z bosons – the electroweak symmetry breaking mechanism. The SM solution is the Higgs mechanism leading to an additional scalar field whose quantum, the Higgs boson, should be experimentally observable [3–5].

Direct searches at the LEP2 experiments ruled out a SM Higgs boson lighter than 114.4 GeV/c^2 (at 95% confidence level). Combining this with direct searches from the Tevatron and indirect constraints from precision electroweak data indicates that the mass of the SM Higgs should be lower than about 158 GeV/c^2 (95% C.L.) [6–8].

Despite its small branching ratio, which reaches no higher than 0.23%, the $\mathrm{H} \to \gamma\gamma$ decay channel provides a clean final-state topology with a mass peak that can be reconstructed with precision. As a consequence, $\mathrm{H} \to \gamma\gamma$ is one of the most promising channels for Higgs discovery in CMS in the very low mass region. Previous searches in this channel have been conducted by the CDF and DØ experiments [9, 10], and also at the LHC by the ATLAS collaboration [11].

A search for a Higgs boson decaying into two photons in pp collisions at a center-of-mass energy of 7 TeV is described below. This search is a development, with additional data, of that reported in Ref. [12].

2 The CMS Detector

The central feature of the CMS detector is a superconducting solenoid, 13 m in length and 6 m in diameter, which provides an axial magnetic field of 3.8 T. The bore of the solenoid is instrumented with particle detection systems. The steel return yoke outside the solenoid is instrumented with gas detectors used to identify muons. Charged particle trajectories are measured by the silicon pixel and strip tracker, with full azimuthal coverage within $|\eta| < 2.5$, where the pseudorapidity η is defined as η = -ln[tan($\theta/2$)], with θ being the polar angle of the trajectory of the particle with respect to the counterclockwise beam direction. A lead-tungstate crystal electromagnetic calorimeter (ECAL) and a brass/scintillator hadron calorimeter (HCAL) surround the tracking volume and cover the region $|\eta| < 3$. In the region $|\eta| < 1.74$, the HCAL cells have widths of 0.087 in both pseudorapidity and azimuth (ϕ). In the (η, ϕ) plane, and for $|\eta| < 1.48$, the HCAL cells map on to 5×5 ECAL crystal arrays to form calorimeter towers projecting radially outwards from points slightly offset from the nominal interaction point. At larger values of $|\eta|$, the size of the towers increases and the matching ECAL arrays contain fewer crystals. A more detailed description of the CMS detector can be found elsewhere [13]. Calibration of the ECAL uses π^0's, $W \to e\nu$ and $Z \to ee$ recorded in 2011. Changes in transparency of the crystals during LHC fills, and subsequent recovery, are monitored continuously and corrected for using injected light from a laser and LED system.

3 Data sample and reconstruction

The dataset used to obtain the results presented here consists of diphoton triggered events corresponding to an integrated luminosity of 4.8 fb^{-1}. Diphoton triggers with asymmetric transverse energy, E_{T}, thresholds and complementary photon selections are used, one selection requiring a loose calorimetric identification based on shower shape and very loose isolation requirements on photon candidates, and the other making only a requirement that the photon

candidate have a high value of the R_9 variable. R_9 is defined as the energy sum of 3×3 crystals centered on the most energetic crystal in the supercluster (see below) divided by the energy of the supercluster, and its value is used in the analysis to reliably distinguish unconverted from converted photons. The E_{T} thresholds are in all cases at least 10% lower than the final selection thresholds. As the instantaneous luminosity delivered by the LHC increased it became necessary to tighten the isolation cut applied in the trigger. To allow for this all 4 possible combinations of threshold and selection criterion were deployed. Taking the logical sum of these triggers results in a trigger efficiency for selected events of almost 100%.

Photon candidates are reconstructed from clusters of ECAL channels around significant energy deposits, which are merged into superclusters. The clustering algorithms result in almost complete recovery of the energy of photons that convert in the material in front of the ECAL. In the barrel region superclusters are formed from 5-crystals-wide strips in η centred on the locally most energetic crystal (seed) and have a variable extension in ϕ. In the endcap, where the ECAL crystals do not have an $\eta \times \phi$ geometry, matrices of 5×5 crystals (which may partially overlap) around the most energetic crystals are merged if they lie within a narrow road in η.

The photon energy is computed starting from the raw supercluster energy. In the endcap the preshower energy is added where the preshower is present. In order to obtain the best resolution, the raw energy must be corrected for the containment of the shower in the clustered crystals, and the shower losses for photons which convert in material upstream of the calorimeter. These corrections are computed using a multivariate regression technique based on the TMVA Boosted Decision Tree implementation [14]. The regression is trained on photons in a Monte Carlo sample using the ratio of the true photon energy to the raw energy as the target variable. The input variables are the global η and ϕ coordinates of the supercluster, a collection of shower shape variables, and a set of local cluster coordinates.

4 Vertex location

The mean number of pp interactions per bunch crossing is about 9.5. The interaction vertices reconstructed from the charged tracks are measured to have an RMS spread in the longitudinal dimension of about 6 cm. If the interaction point is known to better than about 10 mm then the resolution on the opening angle between the photons makes a negligible contribution to the mass resolution, as compared to the ECAL energy resolution. So the mass resolution can be preserved by correctly assigning the reconstructed photons to one of the interaction vertices reconstructed from the charged tracks.

Firstly, the reconstructed primary vertex which most probably corresponds to the Higgs boson interaction can be identified using the kinematic properties of the tracks associated with the vertex and their correlation with the diphoton kinematics. Secondly, if either of the photons converts and conversion tracks are reconstructed and identified, the direction of the converted photon, determined by combining the conversion vertex position and the position of the ECAL supercluster, can be used to point to and so identify the Higgs boson interaction.

For the determination using kinematic properties, 3 discriminating variables are constructed from the tracks associated with each vertex: $\sum p_{\mathrm{T}}^2$, and 2 variables which quantify the p_{T} balance with respect to the diphoton system $-\sum(\vec{p}_T \cdot \frac{\vec{p}_T^{\gamma\gamma}}{|\vec{p}_T^{\gamma\gamma}|})$ and $(\sum p_{\mathrm{T}} - p_{\mathrm{T}}^{\gamma\gamma})/(\sum p_{\mathrm{T}} + p_{\mathrm{T}}^{\gamma\gamma})$. These variables together with an estimate of the pull to each vertex from the longitudinal, z, location on the beam axis pointed to by any recontructed conversion tracks associated with the two photon candidates, $|z_{conversion} - z_{vertex}|/\sigma_{conversion}$, are used in a multivariate system based on

boosted decision trees, to choose the reconstructed vertex to associate with the photons.

The vertex finding efficiency using the kinematic properties of the tracks has been studied with $Z \to \mu\mu$ events where the algorithm is run after the removal of the muons. The use of conversion tracks to locate the vertex is studied with γ+ jet events. In both cases the ratio of the efficiency found in data to that found in Monte Carlo is close to 1. The value is measured as a function of the boson p_{T} and is used as a correction to the Higgs signal model. An uncertainty of 0.4% is ascribed to knowledge of the vertex finding efficiency coming from the statistical uncertainty in the efficiency measurement from $Z \to \mu\mu$ (0.2%) and the uncertainty related to the Higgs p_{T} description which is estimated to be 0.3%. The overall efficiency integrated over the entire p_{T} spectrum for a Higgs boson of mass 120 GeV/c^2 is computed to be 83.0%$\pm$0.2%(stat)$\pm$0.4%(syst) after correction for the difference between data and Monte Carlo.

5 Photon selection

The event selection requires two photon candidates with $p_{\mathrm{T}}^{\gamma 1} > m_{\gamma\gamma}/3$ and $p_{\mathrm{T}}^{\gamma 2} > m_{\gamma\gamma}/4$ within the ECAL fiducial region: $|\eta| < 2.5$ and excluding the barrel-endcap transition region $1.4442 < |\eta| < 1.566$. The fiducial region cut is applied to the supercluster position in the ECAL, and the p_{T} threshold is applied after the vertex assignment. The excluded barrel-endcap transition region excludes the last two rings of crystals in the barrel, to ensure complete containment of accepted showers, and the first ring of triggers tower in the endcap which is obscured by cables and services exiting between the barrel and endcap.

The dominant backgrounds to $\mathrm{H} \to \gamma\gamma$ consist of 1) the irreducible background coming from the prompt two photon production, and 2) the reducible backgrounds from $\mathrm{pp} \to jet + \gamma$ and $\mathrm{pp} \to jet + jet$ where one or more of the photons is non-prompt ("fake"). Photon identification cuts are used to greatly reduce the contributions from from $\mathrm{pp} \to jet + \gamma$ and $\mathrm{pp} \to jet + jet$.

Isolation is a powerful tool to reject the non-prompt background due to electromagnetic showers originating in jets – mainly due to single and multiple π^0's. The multiple collisions in each bunch crossing result in two complications. Firstly, the E_{T} summed in the isolation region in the ECAL and in the hadronic calorimeter (HCAL) includes a contribution from other collisions in the same bunch crossing ("pileup"). The isolation sums in the ECAL and HCAL, and hence both the efficiency and rejection power of a cut on the sums, are then dependent on the number of interactions in the bunch crossing. Secondly, if the vertex is wrongly assigned, the track isolation which requires that the tracks used in the isolation sum are matched to the chosen vertex (so that the sum does not suffer from pileup), will give an isolation sum unrelated to the true isolation of the candidate. This allows non-prompt candidates which are not, in fact, isolated from tracks originating from their interaction point to appear isolated.

The first problem is dealt with by calculating the average transverse energy density in the event, ρ, in regions of the detector separated from the jets and subtracting an appropriate amount, proportional to ρ, from the isolation sums. The second problem is dealt with by applying a cut not only on the isolation sum calculated using the chosen two photon vertex, but also on the isolation sum calculated using the vertex choice hypothesis which maximizes it. The isolation cuts are applied as a constant fraction of the candidate photon E_{T}, effectively cutting harder on low E_{T} photons. It has been shown with $Z \to ee$ events that the resulting variation of selection efficiency with E_{T} is well modelled by the Monte Carlo.

In addition to isolation variables, the following additional variables are used: H/E, the ratio of hadronic energy behind the photon to the photon energy, $\sigma_{i\eta i\eta}$, the transverse width of the

electromagnetic shower, and an electron track veto.

Photon candidates with high values of R_9 are mostly unconverted and have less background than those with lower values. Photon candidates in the barrel have less background than those in the endcap. For this reason it has been found useful to divide photon candidates into 4 categories and apply a different tightness of selection in each category.

The efficiency of the photon identification is measured in data using tag and probe techniques. $Z \to ee$ events are used to determine the efficiency of the complete selection excluding the electron veto cut. Table 1 shows the results for data and Monte Carlo, and the ratio $\epsilon_{data}/\epsilon_{MC}$. The efficiency for photons to pass the electron veto, which approaches 100% in all except category 4, where it is 93%, has been measured using $Z \to \mu\mu\gamma$ events, where the photon is produced by final state radiation, which provide a rather pure source of prompt photons. The ratio $\epsilon_{data}/\epsilon_{MC}$ is close to 1.0 in all categories. The quadratic sum of the statistical and systematic uncertainties for the data measurements of efficiencies are propagated to the uncertainties on the ratios. The ratios are used as corrections to the signal efficiency simulated in the Monte Carlo model of the signal. The uncertainties on the ratios are taken as a systematic uncertainties in the limit setting.

Table 1: Photon identification efficiencies measured in the 4 photon categories using tag and probe with $Z \to ee$ events (for all cuts except electron rejection).

Category	ϵ_{data} (%)	ϵ_{MC} (%)	$\epsilon_{\mathrm{data}}/\epsilon_{\mathrm{MC}}$
Barrel, $R_9 > 0.94$	89.26±0.06±0.04	90.61±0.05	0.985±0.001
Barrel, $R_9 < 0.94$	68.31±0.06±0.55	68.16±0.05	1.002±0.008
Endcap, $R_9 > 0.94$	73.65±0.14±0.39	73.45±0.12	1.002±0.006
Endcap, $R_9 < 0.94$	51.25±0.11±1.25	48.70±0.08	1.052±0.026

The efficiency of the trigger has also been measured using $Z \to ee$ events, with the events classified as described below. For events passing the analysis selection cuts the trigger efficiency is found to be 100% in the high R_9 event classes, and about 99% in the other two classes.

6 Event classes

Search sensitivity can be increased by subdividing the selected events into classes according to indicators of mass resolution and signal to background ratio and combining the results of a search in each class.

Two classifiers are used: the minimum R_9 of the two photons, and the maximum pseudorapidity of the two photons. The class boundary values for R_9 and pseudorapidity are chosen to match those used to categorize photon candidates for the photon identification cuts.

These classifiers are effective in separating diphotons whose mass is reconstructed with good resolution from those whose mass is less well measured, and in separating events with a higher signal to background probability from those with a lower signal to background probability. The fraction of events in each of the 4 classes is shown in Table 2, for signal Monte Carlo, and for data. A Higgs boson of mass 120 GeV/c^2 is chosen for the signal, and the data is counted in a narrow bin ($\pm$ 10 GeV/c^2) centred at 120 GeV/c^2. The table also shows the mass resolution, parameterized both as σ_{eff}: half the width of the narrowest window containing 68.3% of the distribution, and as FWHM/2.35.

Table 2: Fraction of selected events in different event classes, for signal (m_H=120 GeV/c^2) and for data in a narrow bin centred at 120 GeV/c^2. The mass resolution on the signal is also given.

	Both photons in barrel		One or more in endcap	
	$min(R_9)$ >0.94	$min(R_9)$ <0.94	$min(R_9)$ >0.94	$min(R_9)$ <0.94
Signal	31.1%	40.3%	12.2%	16.4%
Data	23.0%	33.8%	17.8%	25.4%
σ_{eff} (GeV/c^2)	1.38	1.84	2.80	3.20
(FWHM/2.35)/m_H	0.99%	1.32%	2.18%	2.55%

7 Signal and background modelling

Monte Carlo simulation of the background processes is not used in the analysis. However the two photon mass spectrum that is observed after the full event selection is found to agree with the distribution predicted by Monte Carlo, within the uncertainties on the cross sections of the contributing processes which is estimated to be about 15%. The background components have been scaled by K-factors obtained from CMS measurements [15–17]. The contribution to the background in the two photon mass range $110 < m_{\gamma\gamma} < 150$ GeV/c^2 from processes giving non-prompt photons is about 30%.

For the analysis the background model is obtained by fitting the observed two photon mass distributions in each of the 4 event classes with 5th order Bernstein polynomials over the range $100 < m_{\gamma\gamma} < 180$ GeV/c^2. The choice of fit function was made based on a study of the bias which might be introduced by the choice of fit function on both the limit in the case of no signal, and the measured signal strength in the case of a signal.

The study of bias was performed as follows. Four functions are fitted to the Monte Carlo background distribution of the two photon invariant mass spectrum and taken as "truth models" for the background. The choice of functions is intended to be sufficienty broad to span the possibilities of the true distribution (i.e. the distribution to which signal-free data would tend if the integrated luminosity were increased). Due to the variation of photon efficiency with E_{T}, and possible kinematic biases imposed by our event classification we are not, at present, able to exclude the possibility that the true distribution has a polynomial form. The 4 functions are:

1. The sum of 2 exponential functions
2. The sum of 2 power functions ($fm^{-p_1} + (1-f)m^{-p_2}$)
3. 2nd order Chebyshev polynomial
4. Laurent series with 4 terms ($f_1m^{-3} + f_2m^{-4} + f_3m^{-5} + (1-f_1-f_2-f_3)m^{-6}$)

Pseudo-experiments describing possible experimental outcomes are then randomly generated using the fitted functions as generators of background. Further pseudo-experiments are generated where a Standard Model signal is added to the background. Exclusion limits are calculated for these experiments using the asymptotic approximation to the CLs procedure [18], using a range of functions to fit the background in the generated experiments. Background fitting functions in shortened ("sliding-window") and extended fit ranges are also tested. The bias is computed as the average value of the observed signal strength in 2000 pseudo-experiments, at each of 9 mass values between 110 and 150 GeV/c^2 (5 GeV/c^2 steps).

It was found that a number of functions were able to reliably predict the local expected background distribution, keeping the bias on the prediction to a small value for all the truth mod-

els. Among these functions it was found that a 5th order polynomial fit to the range $100 < m_{\gamma\gamma} < 180$ GeV/c^2 best preserved the sensitivity of the analysis to excluding or finding a Higgs Boson signal. The maximum average bias found when using the 5th order polynomial fit was compared to the statistical uncertainty in the background fit, evaluated as an absolute number of events in a 3 GeV/c^2 window. It is found that the statistical uncertainties are at least 5 times larger than the remaining bias at all of the 9 mass points.

The description of the Higgs boson signal used in the search is obtained from Monte Carlo simulation using the NLO matrix-element generator POWHEG [19, 20] interfaced with PYTHIA. The Higgs boson transverse momentum spectrum has been reweighted to the NNLL+NLO distribution computed by the HqT program [21, 22]. The simulation includes an accurate simulation of the distribution of the number of interactions taking place in each bunch crossing.

Detailed study of the mass distribution of $Z \to ee$ events and comparison with the Monte Carlo simulation indicates that corrections need to be made to the measured energy of the photons. This fine-tuning of the calibration is applied to electrons and photons with different values in 6 separate subsets of the dataset corresponding to periods with different derivation and treatment of the crystal transparency corrections. After the application of these corrections the $Z \to ee$ events are re-examined and values are derived for the random smearing that needs to be made to the Monte Carlo simulation to account for the energy resolution observed in the data. These smearings are derived for photons separated into 4 η regions, 2 in the barrel and 2 in the endcap, and 2 categories of R_9. The uncertainties on the measurements are taken as systematic uncertainties in the limit setting.

The $m_{\gamma\gamma}$ distribution for the combined data in all 4 event classes together with the weighted sum of fits to the individual event classes, is shown in Fig. 1. The uncertainty bands are computed from the fit uncertainty on the background yield within each bin used for the data points. The figure also shows the expected signal shape (for $m_H = 120\,\text{GeV}/c^2$). The magnitude of the signal is what would be expected if its cross section were 5 times the SM expectation.

8 Results

The confidence level (CL) for exclusion or discovery is evaluated using the two photon invariant mass distribution as the observable for each of the event classes previously defined. The results in the 4 classes are combined in the CL calculation to obtain the final result.

Two statistical approaches are considered in evaluating limits: a modified frequentist approach (CL_S) using the profile likelihood as a test statistic [18], and a Bayesian approach with a flat prior for the signal strength. These two methods are generally expected to give similar results and so provide a valuable cross check of the statistical procedures.

Both a binned and an unbinned evaluation of the likelihood are considered. While most of the analysis and determination of systematic uncertainties are common for these two approaches, there are differences at the final stages which make a comparison useful. The signal model is taken from the MC after applying the corrections determined from data/Monte Carlo comparisons of $Z \to ee$ and $Z \to \mu\mu\gamma$ mentioned above, and the reweighting of the p_{T}^{H} spectrum. In the unbinned evaluation the signal model is parametric, based on analytic functions fitted to the Monte Carlo, whereas the binned evaluation uses templates made with Monte Carlo events. The comparison of results thus verifies that the parametric model describes the Monte Carlo well. For the background, Monte Carlo is not used and the background is evaluated from a fit to the data.

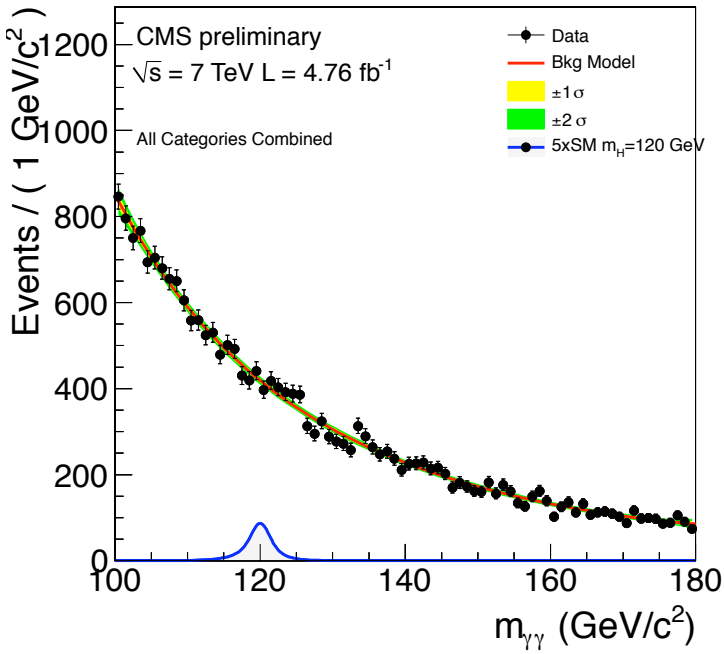

Figure 1: Background model fit to the $m_{\gamma\gamma}$ distribution for the combined data in all 4 event classes, together with a simulated signal (m_H=120 GeV/c^2). The magnitude of the signal is what would be expected if its cross section were 5 times the SM expectation.

Given the narrowness of the Higgs mass peak which has a resolution approaching 1 GeV/c^2 in the classes with best resolution, the search is carried out in steps of 0.5 GeV/c^2.

Table 3 lists the sources of systematic uncertainty that have been taken into account in the evaluation of the limits, together with the magnitude of the variation of the source that has been applied.

The limit set on the cross section of a Higgs boson decaying to two photons using the frequentist CL_S computation and an unbinned evaluation of the likelihood, is shown in Fig. 2. Figure 3 shows the limit relative to the SM expectation, where the theoretical uncertainties on the expected cross sections from the different production mechanisms are individually included as systematic uncertainties in the limit setting procedure. The fluctuations of the observed limit about the expected limit are consistent with statistical fluctuations to be expected in scanning the mass range. It has also been verified that the shape of the observed limit obtained is unchanged if the choice of background model fitting function is changed over a wide range of functional forms, although the expected limit improves by as much as 10% if functions with less free parameters than the 5th order polynomial are used.

The results obtained from the binned evaluation of the likelihood are in excellent agreement with the results shown in Figs. 2 and 3.

Figure 4 shows the local p-value calculated, using the asymptotic approximation, at 0.5 GeV/c^2 intervals in the mass range $110 < m_H < 150$ GeV/c^2. The local p-value quantifies the probability for the background to produce a fluctuation as large as observed, and assumes that the relative signal strength between the event classes follows the Monte Carlo signal model for the Standard Model. The local p-value corresponding to the largest upwards fluctuation of the observed limit, at 123.5 GeV/c^2, has been computed to be 9.6×10^{-3} (2.34σ) in the asymptotic

Table 3: Separate sources of systematic uncertainties accounted for in the limit setting. The magnitude of the variation of the source that has been applied is shown in the righthand column.

Source		**Uncertainty**	
Photon identification efficiency:	barrel	1.0%	
	endcap	2.6%	
R_9 >0.94 efficiency	barrel	4%	
(results in class migration)	endcap	6.5%	
		$R_9 > 0.94$	$R_9 < 0.94$
Energy resolution ($\Delta\sigma/E_{MC}$):	barrel low η, high η	0.22%, 0.61%	0.24%, 0.59%
	endcap low η, high η	0.91%, 0.34%	0.30%, 0.53%
Energy scale ($(E_{data} - E_{MC})/E_{MC}$)	barrel low η, high η	0.19%, 0.71%	0.13%, 0.51%
	endcap low η, high η	0.88%, 0.19%	0.18%, 0.28%
Integrated luminosity		4.5%	
Trigger efficiency: One or more photons $R_9 < 0.94$ in endcap		0.4%	
	Other events	0.1%	
Vertex finding efficiency		0.4%	
gluon fusion process cross section (scale)		+12.5% -8.2%	
gluon fusion process cross section (PDF)		+7.9% -7.7%	
Vector boson fusion process cross section (scale)		+0.5% -0.3%	
Vector boson fusion process cross section (PDF)		+2.7% -2.1%	
Associated production with W/Z cross section (scale)		+1.8% -1.8%	
Associated production with W/Z cross section (PDF)		+4.2% -4.2%	
Associated production with $t\bar{t}$ cross section (scale)		+3.6% -9.5%	
Associated production with $t\bar{t}$ cross section (PDF)		+8.5% -8.5%	

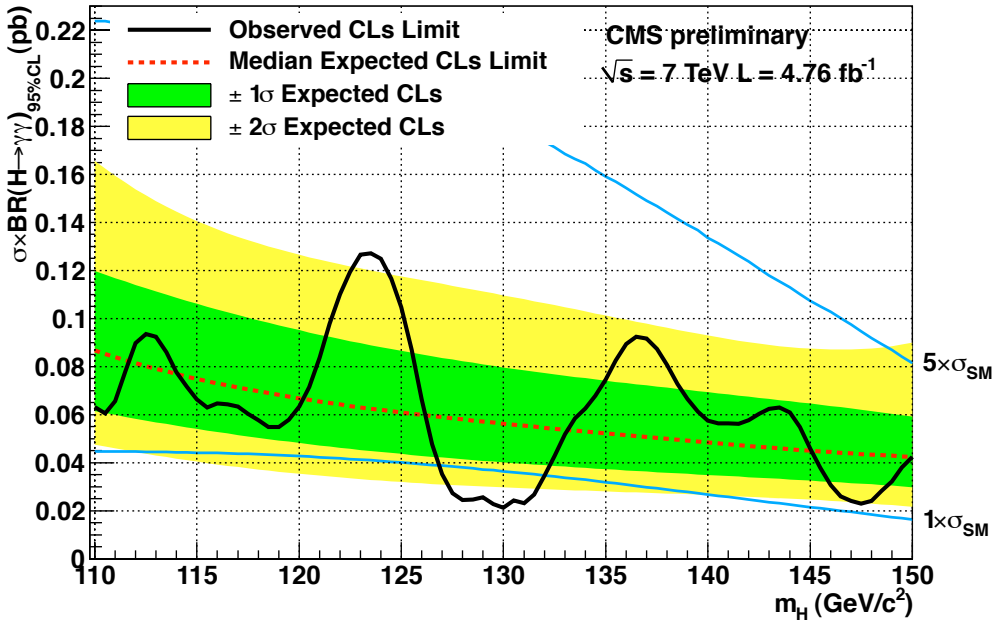

Figure 2: Exclusion limit on the cross section of a SM Higgs boson decaying into two photons as a function of the boson mass.

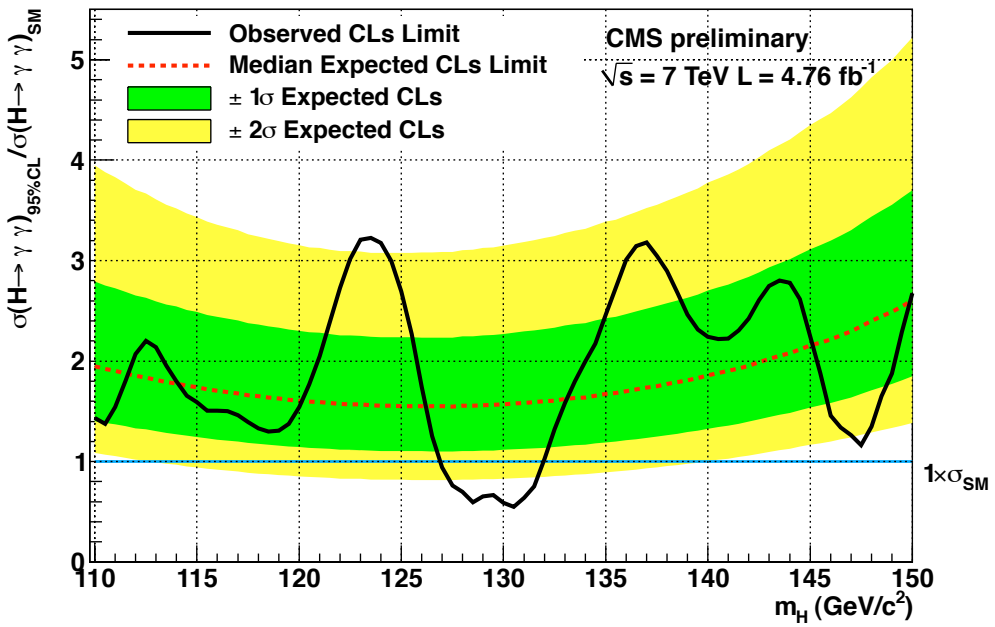

Figure 3: Exclusion limit on cross section of a SM Higgs boson decaying into two photons relative to the SM cross section, as a function of the boson mass.

approximation, and 10.5×10^{-3} (2.31σ) when the calculation uses pseudo-data (the pseudo-data value is shown as a blue point in Fig. 4). Since a fluctuation of the background could occur at any point in the mass range there is a look-elsewhere effect. When this is taken into account the probability, under the background only hypothesis, of observing a similar or larger excess in the full range is 0.21 ± 0.01 (0.79σ). Three of the 4 event classes contribute to the excess at 123.5 GeV/c^2, and overall the fitted signal strength is 1.7 ± 0.8 times the Standard Model cross section.

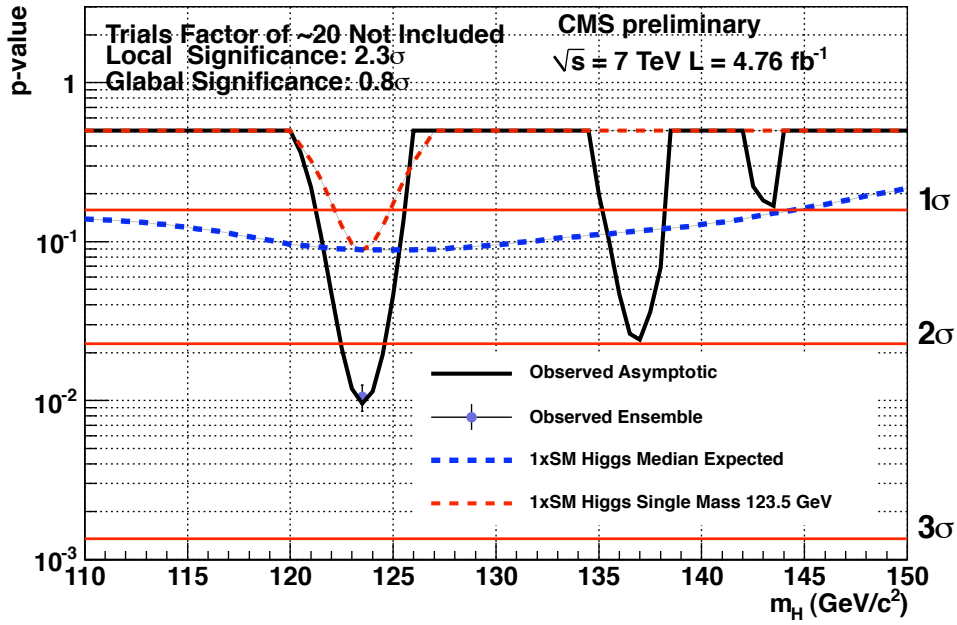

Figure 4: The p-value, which measures the probability that the observed events are compatible with a background only hypothesis.

9 Summary

A search has been made for a Higgs boson decaying into two photons in the CMS detector. The analysis uses reconstruction and selection techniques that take account of the multiple interactions occurring in each bunch crossing in the current LHC operating conditions, which affect isolation and identification of the interaction vertex. The selected events are subdivided into classes according to indicators of mass resolution and signal to background ratio, and the results of a search in each class are combined. The expected exclusion limit at 95% CL is between 1.5 and 2.0 times the Standard Model cross section in the mass range between 110 and 140 GeV/c^2, and the observed limit disfavours at 95% CL a Standard Model Higgs boson decaying into two photons in the mass range 127 to 131 GeV/c^2.

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